

Physics 5A Discussion (Week 0)

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Office hour: Thursday 9-10 AM at PAB 2-429

Example: We know 1 foot is 0.305 meters. Denote “ft” for feet and “m” for meters, so

$$100 \text{ ft}^2 = (10 \text{ feet}) \cdot (10 \text{ feet}) = (3.05 \text{ meters}) \cdot (3.05 \text{ meters}) = 9.30 \text{ m}^2.$$

Problem 1: The radius of a disc is 5 meters. The area of a disc is $A = \pi r^2$ where r is the radius. What is its area in square feet (ft^2)?

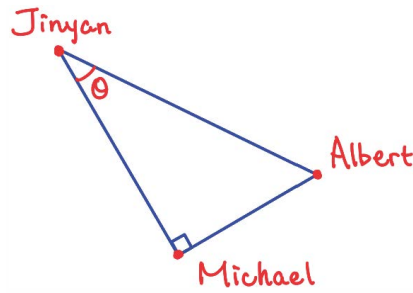
Example: If the force F acting on an object of mass m (measured in kilograms) and acceleration a (measured in meters per second squared) obeys the formula $F = ma$, then the unit of the force is

$$[F] = [m] \cdot [a] = (\text{kilogram}) \cdot (\text{meters per second squared}) = \text{kg m} / \text{s}^2.$$

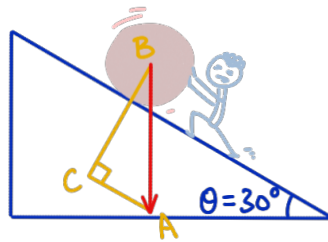
Later on we will call this new unit (kg m/s^2) as Newton (N). The letter m is italicized when we use it to denote mass, but m is not italicized when we use it to denote the unit meter.

Problem 2: Knowing that force F is measured in kg m/s^2 , density ρ is measured in kg/m^3 , and velocity v is measured in m/s , find the unit of the quantity B in the formula

$$F = \rho v^2 B.$$



Problem 3: (Make sure you include units in your final answers!) Using the figure shown, if the distance between Jinyan and Michael is 4 meters, and the distance between Michael and Albert is 3 meters, what is the distance between Jinyan and Albert? What is the angle θ in degrees? What is the angle θ in radian?



Problem 4: Jinyan is pushing a huge rock up an incline, which makes an angle 30° above the horizontal. His friend Michael is trying to analyze the force that Jinyan needs to hold the rock, where Michael needs to analyze the geometry drawn in the figure: There is a small right triangle $\triangle ABC$. The line segment AC is parallel to the incline, and the line segment BC is perpendicular to the incline. If the force arrow BA has a “length” of 100 Newtons, find the “lengths” of the line segments AC and BC . All these “lengths” are measured in Newtons, so your final answer should be expressed with the unit N. (*Hint:* First find out the angle of $\angle ABC$ using the given information, then use the sin and cos functions to find the length of the edges of the right triangle.)

Problem 1

I recommend first convert everything to the unit that you want, then do the algebra and computations, so we first we do

$$r = 5 \text{ m} = \frac{5}{0.305} \text{ ft},$$

then we calculate the area

$$A = \pi r^2 = \pi \cdot \left(\frac{5}{0.305} \text{ ft} \right)^2 = \pi \cdot \left(\frac{5}{0.305} \right)^2 \text{ ft}^2 \approx 844 \text{ ft}^2.$$

Make sure you have the correct unit in your final answer. For this class, when it comes to decimal places, I recommend keeping 3 or 4 significant digits unless specified otherwise.

Problem 2

We want to find the unit of B , and the units of all other quantities are known, so we rearrange the $F = \rho v^2 B$ formula to isolate B ,

$$B = \frac{F}{\rho v^2},$$

then we just plug in the units and get

$$[B] = \frac{\text{kg m/s}^2}{(\text{kg/m}^3) \cdot (\text{m/s})^2} = \text{m}^2.$$

Now we see that B has the unit of m^2 , which is something like an “Area”.